## ANSWER KEY & MARKING SCHEME SET 1

**Full Syllabus (Chapters 1–13)** | Class 8 Science | Maximum Marks: 80

### SECTION — A (Objective Type) — Answers — 20 × 1 = 20

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains why one factor is changed. |
| 2 | **(c)** A is true but R is false. | A is true; R is false because animal cells lack a cell wall. |
| 3 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 4 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 5 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 6 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 7 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 8 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 9 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 10 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 11 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 12 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 13 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 14 | **(d)** greater | 12 degrees C versus 7 degrees C. |
| 15 | **(d)** 2 mm | 2 x 1000 = 2000 micrometres = 2 mm. |
| 16 | **(a)** 70 | 40 + 30 = 70. |
| 17 | **(d)** B gets hotter than A | Thin wire has higher resistance. |
| 18 | **(d)** 20 N | 60 - 40. |
| 19 | **(d)** 3822 Pa | Area = 3.14 x 0.01 = 0.0314 square m; 120/0.0314 = 3822 Pa. |
| 20 | **(b)** spaces between particles | Particles of one fill the gaps. |

### SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | T = 34 / 20 = 1.7 s (1); Timing many swings reduces the error of reaction time (1). | 1+1 |
| 22 | Actual = 5 / 500 = 0.01 mm (1); = 10 micrometres (1). | 1+1 |
| 23 | Unvaccinated = 2000 = 25% (1); Fewer susceptible persons means less spread (1). | 1+1 |
| 24 | Both feet are at the same potential (1); No current passes through the bird (1). | 1+1 |
| 25 | Mass remains 3 kg (1); Weight = 30 / 6 = 5 N (1). | 1+1 |
| 26 | Area = 0.005 square metre (1); P = 50/0.005 = 10000 Pa (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 27 | Only one soil type was compared (1); The conclusion over-generalises (1); Test several soils with controls (1). | 1+1+1 |

| | | |
|---|---|---|
| **28** | Onion cells are rectangular with a cell wall; cheek cells are rounded or irregular without a wall (1); Both show a nucleus and cytoplasm; the onion cell has a large vacuole (1); Stains such as safranin or methylene blue make the parts visible (1). | 1+1+1 |
| **29** | Contaminated drinking water (1); Unhygienic food (1); Preventive: boiled water, hygiene, treat water tank (1). | 1+1+1 |
| **30** | Increase the number of turns (1); Increase the number of cells (1); Use a thicker soft iron core (1). | 1+1+1 |
| **31** | Friction = 30 N (1); With 20 N net force = 10 N backwards (1); The box slows down and stops (1). | 1+1+1 |
| | **OR:** Area = 0.01 square m; P = 50/0.01 = 5000 Pa (1); Stacked: weight 100 N (1); P = 10000 Pa (1). | 1+1+1 |
| **32** | -10 to 0: ice warms (1); 0 to 100: melting at 0 then water warms (1); At 100: boiling then steam warms to 110 (1). | 1+1+1 |
| | **OR:** Probably iron, an element (1); Magnetic and reacts with acid giving hydrogen (1); Could also be a mixture containing iron (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers

**6 × 5 = 30**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **33** | Apparatus: equal pieces of cloth, beaker, balance (1); Independent: type of fabric (1); Constants: size, water volume, soaking time (1); Measure mass before and after, tabulate gain (1); Fabric with greatest gain absorbs most (1). | 1+1+1+1+1 |
| **34** | Onion peel stained with safranin (1); Cheek cells stained with methylene blue (1); Plant: rectangular, wall, large vacuole (1); Animal: rounded, irregular, no wall (1); Both: nucleus and cytoplasm (1). | 1+1+1+1+1 |
| **35** | Contaminated food or water at the feast (1); Salmonella bacteria from a carrier or dirty water (1); Hands and flies spread germs (1); Ensure clean water, cooked food, hygiene (1); Treatment with antibiotics on advice; vaccination (1). | 1+1+1+1+1 |
| **36** | Independent: number of turns; dependent: pins lifted (1); Constant: cell, nail, wire (1); Procedure: 10, 20, 30, 40 turns and count pins (1); Expected: pins rise with turns (1); Conclusion (1). | 1+1+1+1+1 |
| | **OR:** Independent: weight on block (1); Dependent: pull on spring balance at constant speed (1); Constant: surface and speed (1); Table of load and friction (1); Graph is a straight line rising (1). | 1+1+1+1+1 |
| **37** | Pressure at 3 m is three times that at 1 m (1); Ears feel more pressure at depth (1); Dam base is thick (1); Pressure acts in all directions (1); Independent of container shape (1). | 1+1+1+1+1 |
| | **OR:** Independent: temperature of water (1); Dependent: time taken for ink to spread (1); Constants: volume, ink drop, beaker (1); Observation: spreads faster in warm water (1); Conclusion: higher temperature gives faster diffusion (1). | 1+1+1+1+1 |
| **38** | Prepare a mixture of iron and sulphur (1); Test with a magnet before heating (1); Heat strongly in a test tube (1); Black, non-magnetic product forms after heating (1); A compound is formed; mixtures retain properties (1). | 1+1+1+1+1 |
| | **OR:** At 70: 150 g solution = 50 g salt + 100 g water (1); At 20 the 100 g water holds 20 g (1); Crystals = 50 - 20 (1); = 30 g (1); Percentage of solution at 20: 20/120 = 16.7 per cent (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## ANSWER KEY & MARKING SCHEME SET 2

**Full Syllabus (Chapters 1–13)** | Class 8 Science | Maximum Marks: 80

### SECTION — A (Objective Type) — Answers — 20 × 1 = 20

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 36 g, 1 : 8 | 4 + 32 = 36; 4:32 = 1:8. |
| 2 | **(b)** 39 per cent | 64/164 x 100 = 39. |
| 3 | **(d)** 140 degrees | Initial i = 55; new i = 70; angle = 140. |
| 4 | **(c)** 15 m | 3/4 = h/20. |
| 5 | **(b)** snake | Energy decreases at each step. |
| 6 | **(a)** 0.53 | 3400/6400. |
| 7 | **(b)** 50 degrees C | Subtract zero error of 3. |
| 8 | **(a)** stops functioning properly | Nucleus controls the cell. |
| 9 | **(d)** colds are viral and often recover by themselves | Colds go away without them. |
| 10 | **(c)** the iron rod | Iron concentrates the field. |
| 11 | **(d)** moves towards team A | Net force towards A. |
| 12 | **(a)** A is likely to be stronger | Sea supplies energy. |
| 13 | **(c)** temperature increases the motion of particles | Faster particles. |
| 14 | **(d)** homogeneous mixture | Alloy. |
| 15 | **(b)** Y | Immiscible liquids. |
| 16 | **(d)** 90 degrees to the surface (back along normal) | Incidence along the normal. |
| 17 | **(c)** about 400 times farther | Equal angular sizes. |
| 18 | **(d)** bees and flowers | Both benefit. |
| 19 | **(a)** flooding by rising sea levels | Higher sea level. |
| 20 | **(c)** A is true but R is false. | A is true; R is false, repetition checks reliability but cannot guarantee truth. |

### SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Gas particles move rapidly in all directions (1); They spread uniformly and fill the container (1). | 1+1 |
| 22 | Oxygen = 36 - 4 = 32 g (1); Ratio = 4 : 32 = 1 : 8 (1). | 1+1 |
| 23 | Percentage = 37.5 per cent (1); Water = 62.5 g (1). | 1+1 |
| 24 | Initial i = 60 degrees (1); New i = 70 or 50 degrees (1). | 1+1 |
| 25 | 365.25 is slightly more than the actual 365.2422 (1); 1900 is not a leap year but 2000 is (1). | 1+1 |
| 26 | Grass is the producer; deer the primary consumer; tiger the secondary consumer (top carnivore) (1); The tiger level has the least energy (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 27 | NaCl: 4 g undissolved (1); KNO3: 8 g undissolved (1); NaCl is more soluble (1). | 1+1+1 |
| 28 | n = 360/theta - 1 = 7 (1); 360/theta = 8 (1); theta = 45 degrees (1). | 1+1+1 |
| 29 | At noon: almost no shadow (1); At 3 pm: a shadow pointing east (1); Length is larger than at noon (1). | 1+1+1 |
| 30 | Energy decreases at each trophic level (1); A small amount of energy reaches the top level (1); So only a few top carnivores can be supported (1). | 1+1+1 |
| 31 | Dense iron sank to the centre (1); Lighter material floated up (1); Seismic S-waves do not pass through the outer core (1). | 1+1+1 |
| | **OR:** Straight line (1); Passes through the origin (1); Ratio stays constant (1). | 1+1+1 |
| 32 | Digestion: food vacuole (1); Excretion: contractile vacuole (1); Movement: pseudopodia (1). | 1+1+1 |

| | | |
|---|---|---|
| | **OR:** Goitre: iodised salt (1); Rickets: milk, sunlight (1); Anaemia: leafy vegetables (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers

**6 × 5 = 30**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **33** | f = R/2 (1); = 20/2 = 10 cm (1); Sun's rays meet at the principal focus, 10 cm from the pole (1); Bulb is placed at the focus (1); Reflected rays form a parallel beam (1). | 1+1+1+1+1 |
| **34** | Measure the shadow of a vertical stick at local noon each week (1); Longest in December (1); Shortest in June (1); Medium at equinoxes (1); Graph shows a smooth annual cycle (1). | 1+1+1+1+1 |
| **35** | Herbivores increase (1); Overgrazing of grass (1); Soil erosion (1); Water and shelter loss (1); Ecosystem degrades; other species decline (1). | 1+1+1+1+1 |
| **36** | Inner core = 6400 - (35 + 2900 + 2200) (1); = 1265 km (1); Crust 0.5 per cent, mantle 45 per cent (1); Outer core 34 per cent, inner core 20 per cent (1); Mantle is the thickest (1). | 1+1+1+1+1 |
| | **OR:** Curiosity starts the questions (1); Careful observation gives evidence (1); Persistence helps overcome failed trials (1); Repetition ensures reliability (1); Results shared for checking (1). | 1+1+1+1+1 |
| **37** | Cells in a multicellular body divide the work among themselves (1); Nerve cells are long and branched to carry messages (1); Red blood cells carry oxygen (1); Root hair cells have long projections to absorb water (1); Specialisation makes the body work efficiently (1). | 1+1+1+1+1 |
| | **OR:** Pathogen causes disease; host harbours it (1); Vector carries it (1); Symptoms felt, immunity resists (1); Vaccine prevents; antibiotic treats bacteria (1); Epidemic is a sudden spread (1). | 1+1+1+1+1 |
| **38** | Cell about 1.5 V (1); Battery: several cells, higher voltage (1); Mains about 220 V (1); Low voltage cannot drive a harmful current through the body (1); Mains can cause fatal shock (1). | 1+1+1+1+1 |
| | **OR:** Streamlined shape reduces drag (1); Birds and fish (1); Cars and trains (1); Aeroplanes (1); Boats and bullets (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

## ANSWER KEY & MARKING SCHEME SET 3

**Full Syllabus (Chapters 1–13)** | Class 8 Science | Maximum Marks: 80

### SECTION — A (Objective Type) — Answers — 20 × 1 = 20

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** Both A and R are true and R is the correct explanation of A. | Chloroplasts carry out photosynthesis, which is why plants (which make their own food) have them. |
| 2 | **(a)** Both A and R are true and R is the correct explanation of A. | Antibodies prepare the body. |
| 3 | **(a)** Both A and R are true and R is the correct explanation of A. | Series connection allows it to sense all current. |
| 4 | **(a)** Both A and R are true and R is the correct explanation of A. | Weight equals air resistance. |
| 5 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 6 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 7 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 8 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 9 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 10 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 11 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 12 | **(a)** Both A and R are true and R is the correct explanation of A. | R explains A. |
| 13 | **(b)** B and C | Only light differs between B and C. |
| 14 | **(a)** 0.05 mm | 5 / 100 = 0.05 mm. |
| 15 | **(d)** 75% | 15000/20000 x 100 = 75%. |
| 16 | **(c)** glows dimmer | Reversed cell opposes the others. |
| 17 | **(d)** 200 N | 600/3. |
| 18 | **(a)** 4000 Pa | Inversely proportional. |
| 19 | **(c)** change of state | Latent heat. |
| 20 | **(a)** 1 g and 8 g | Ratio 1 : 8. |

### SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | 0.15 x 1.08 x 10^12 (1); = 1.6 x 10^11 cubic km (1). | 1+1 |
| 22 | Accuracy is closeness to the true value; precision is closeness of repeated readings (1); Readings may be precise but wrong due to zero error (1). | 1+1 |
| 23 | Amoeba has only a flexible membrane (1); Plant cell has a rigid wall (1). | 1+1 |
| 24 | They cannot be cured completely (1); Treatment and lifestyle changes keep them in control (1). | 1+1 |
| 25 | Total resistance increases (1); Less current for each (1). | 1+1 |
| 26 | Friction is lower (1); Less force opposes the sliding (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 27 | Wire resistance is high (1); Less current reaches the bulb (1); Use a short thick wire (1). | 1+1+1 |

| | | |
|---|---|---|
| **28** | Friction depends on the force pressing surfaces (1); Larger area but less pressure per unit area (1); These balance out (1). | 1+1+1 |
| **29** | Pressure gradient is steepest (1); Air rushes toward the centre (1); Rotation increases speed (1). | 1+1+1 |
| **30** | Ice: least energy, close and fixed (1); Water: medium (1); Steam: highest, far apart (1). | 1+1+1 |
| **31** | In diamond all electrons are held in fixed bonds (1); In graphite some electrons are free to move (1); Different arrangements (1). | 1+1+1 |
| | **OR:** Dissolve impure crystals in hot water (1); Filter the hot solution (1); Cool slowly to get pure crystals (1). | 1+1+1 |
| **32** | Close: virtual, erect and enlarged image (1); Far: real and inverted image (1); So the face appears inverted at a distance (1). | 1+1+1 |
| | **OR:** Earth rotates 360 degrees in 24 hours (1); 15 degrees per hour (1); Local noon differs with longitude (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers

**6 × 5 = 30**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **33** | Pushing in reduces volume, pressure rises (1); More collisions with walls (1); Pulling out lowers pressure (1); Outside air enters the pump (1); Pump compresses air into tyre (1). | 1+1+1+1+1 |
| **34** | Increase = 1600 times (1); Particles are far apart in the gas (1); Strong attraction in liquid (1); Little space between liquid particles (1); Gas fills available space (1). | 1+1+1+1+1 |
| **35** | Oxygen: molecules of two oxygen atoms (1); Water: molecules with H and O atoms bonded (1); Air: different molecules mixed (1); Atoms joined in compounds (1); Mixtures are not bonded (1). | 1+1+1+1+1 |
| **36** | Boil solution in a flask (1); Vapour passes through a condenser (1); Pure water collects (1); Salt remains (1); No dissolved impurities (1). | 1+1+1+1+1 |
| | **OR:** Cornea and lens focus light (1); Image forms on retina (1); Myopia: concave lens (1); Hypermetropia: convex lens (1); Cataract and other causes (1). | 1+1+1+1+1 |
| **37** | Latitude = 30 degrees N (1); Equator: on the horizon (1); North Pole: overhead (1); Height increases as he travels north (1); Pole Star altitude equals latitude (1). | 1+1+1+1+1 |
| | **OR:** Global warming (1); Ice melts and sea level rises (1); Ocean acidification (1); Extreme weather (1); Individual: save energy and plant trees (1). | 1+1+1+1+1 |
| **38** | Atmosphere: warming (1); Oceans: more acidic, coral damage (1); Plants: faster growth up to a point (1); But heat and drought reduce crops (1); Net effect is harmful (1). | 1+1+1+1+1 |
| | **OR:** Height increases up to 60 mL (1); Falls at 80 mL (1); Too much water may reduce air in the soil (1); Next test: values between 40 and 80 mL (1); Repeat with several plants (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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